

GitHub Repo File Explainer

Chrome Extension · Development Build Tracker · 10 weeks · 4 phases · 64 tasks

Overview

Phase	Timeline	Effort	Tasks	Deliverable
Phase 1 — Detection & scaffold	Weeks 1–2	~18 hrs	16	Loadable extension that detects any GitHub file page, shows sidebar shell with correct filename/language badge and skeleton state
Phase 2 — File reader & language detection	Weeks 3–5	~28 hrs	16	Extension reliably extracts clean file content for any text-based format on GitHub, with language label, token-safe truncation, and caching
Phase 3 — AI summary engine	Weeks 6–8	~32 hrs	16	Sidebar shows plain-English summary + technical breakdown for any file format, with follow-up Q&A; and streaming render
Phase 4 — Polish & ship	Weeks 9–10	~22 hrs	16	Published Chrome extension supporting GitHub + GitLab, with onboarding, copy actions, dark mode, and token usage visibility

Phase 1 — Detection & scaffold

Weeks 1–2 · ~18 hrs

Deliverable: Loadable extension that detects any GitHub file page, shows sidebar shell with correct filename/language badge and skeleton state

Task group	Task	Owner	Start date	Status
Project scaffold	Manifest v3 skeleton (reuse from TF extension)	Dev		Not started
	Separate repo or monorepo alongside TF explainer	Dev		Not started
	ESLint, Prettier, web-ext hot reload	Dev		Not started
	GitHub Actions CI for lint + build	Dev/CI		Not started
File page detection	Detect github.com/*/*blob/* URL pattern	Dev		Not started
	MutationObserver for SPA navigation between files	Dev		Not started
	Extract filename, extension, repo name from URL + DOM	Dev		Not started
	Skip binary files (images, fonts, archives)	Dev		Not started

Task group	Task	Owner	Start date	Status
Sidebar shell	Inject sidebar alongside GitHub file view	Dev/UX		Not started
	Language badge + file name in header	Dev/UX		Not started
	Skeleton loading state	Dev/UX		Not started
	Collapse / pin toggle	Dev/UX		Not started
Settings popup	Reuse API key storage from TF explainer if same extension	Dev		Not started
	Audience toggle: non-technical vs developer mode	Dev/UX		Not started
	Per-repo opt-out list	Dev		Not started
	Auto-summarise on open vs manual trigger toggle	Dev/UX		Not started

Phase 2 — File reader & language detection			Weeks 3–5 · ~28 hrs	
Deliverable: Extension reliably extracts clean file content for any text-based format on GitHub, with language label, token-safe truncation, and caching				
Task group	Task	Owner	Start date	Status
Content extraction	Read raw file lines from GitHub DOM code table	Dev		Not started
	Fallback: fetch raw.githubusercontent.com URL	Dev		Not started
	Handle very large files — read first 300 + last 50 lines	Dev		Not started
	Strip rendered HTML, keep plain text only	Dev		Not started
Language detection	Detect from file extension (.py, .go, .ts, .yaml, .sql...)	Dev		Not started
	Fallback: read GitHub's own language badge from DOM	Dev		Not started
	Map ~40 common extensions to language names	Dev		Not started
	Flag unsupported / binary types gracefully	Dev		Not started
Token budgeting	Estimate token count (~4 chars per token)	Dev		Not started
	Hard cap: 3,000 tokens of code sent to API	Dev		Not started
	Smart truncation: keep imports + first + last function	Dev		Not started

Task group	Task	Owner	Start date	Status
	Show 'file truncated' notice in sidebar when active	Dev/UX		Not started
Caching layer	Cache key: repo + branch + filepath + last-commit SHA	Dev		Not started
	Store in chrome.storage.local (max 5MB)	Dev		Not started
	Evict LRU entries when approaching storage limit	Dev		Not started
	Show cached badge + timestamp in sidebar	Dev/UX		Not started

Phase 3 — AI summary engine

Weeks 6–8 · ~32 hrs

Deliverable: Sidebar shows plain-English summary + technical breakdown for any file format, with follow-up Q&A; and streaming render

Task group	Task	Owner	Start date	Status
Prompt engine	System prompt tuned for mixed audiences	Dev		Not started
	Non-technical mode: plain English, no jargon, real-world analogy	Dev		Not started
	Developer mode: what it does, dependencies, key functions, gotchas	Dev		Not started
	Language-aware prompt shape for SQL/Python/YAML/Dockerfile	Dev		Not started
Summary cards	'What this file does' — 2-sentence plain summary always first	Dev/UX		Not started
	'Key things inside' — bullet list of functions/classes/config	Dev/UX		Not started
	'How it connects' — what imports/calls this file (inferred)	Dev/UX		Not started
	'Watch out for' — TODOs, complexity, hardcoded values flagged	Dev/UX		Not started
Ask follow-up	Inline text input: 'Ask anything about this file'	Dev/UX		Not started
	Sends file content + prior summary as context	Dev		Not started
	Streaming response rendered inline	Dev		Not started
	Suggested question chips in sidebar	Dev/UX		Not started
Performance	Streaming API response — first words appear in ~1s	Dev		Not started

Task group	Task	Owner	Start date	Status
	Progressive render: plain summary first, cards fill after	Dev		Not started
	Auto-trigger on file open (if enabled) or manual button	Dev		Not started
	Abort controller to cancel in-flight requests on navigation	Dev		Not started

Phase 4 — Polish & ship

Weeks 9–10 · ~22 hrs

Deliverable: Published Chrome extension supporting GitHub + GitLab, with onboarding, copy actions, dark mode, and token usage visibility

Task group	Task	Owner	Start date	Status
GitLab support	Add gitlab.com/*blob/* URL matcher	Dev		Not started
	GitLab-specific DOM selectors for code content	Dev		Not started
	Test across GitLab file view, raw, and blame	Dev/Test		Not started
	Self-hosted GitLab: allow custom domains in settings	Dev		Not started
Copy & export	Copy summary button — pastes plain-text to clipboard	Dev/UX		Not started
	Copy as Markdown — formatted with file name + bullets	Dev/UX		Not started
	Share link: opens Claude.ai with summary pre-filled	Dev/UX		Not started
	Export to Notion / Confluence via URL handler (optional)	Dev		Not started
UX polish	First-install onboarding: API key setup in 3 steps	Dev/UX		Not started
	Keyboard shortcut: Alt+S to trigger summary	Dev		Not started
	Dark mode sidebar matching GitHub dark theme	Dev/UX		Not started
	Token usage meter in popup so users track spend	Dev/UX		Not started
Build & ship	Vite/esbuild bundle, minify for prod	Dev/CI		Not started
	Automated zip for Chrome Web Store submission	Dev/CI		Not started

Task group	Task	Owner	Start date	Status
	Privacy policy (extension reads code — must disclose API usage)	Docs		Not started
	README, CHANGELOG, and store listing copy	Docs		Not started

Status legend

Not started	In progress	Done	Blocked
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